



Towards FAIR Architectural Data Collections

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Keywords: FAIR data, research data management, architectural research data, knowledge graphs, provenance documentation

Abstract

Findable, accessible, interoperable and reusable (FAIR) data (Wilkinson, 2016) is crucial for contemporary research. While infrastructure such as National Research Data Infrastructure (NFDI) in Germany (Schmitt, 2018), Thematic Digital Competence Centres (TDCC) in Netherlands or European Open Science Cloud (EOSC) are emerging, architectural research produces large amounts of data rarely suited as FAIR publication. Tools and practices for aggregating, processing and publishing domain specific data remain insufficient.

Architectural information and data is commonly shared in two ways:

1. **Within design and planning**, to support project teams and specialists in the design and construction process, e.g. Building Information Models (BIM) stored in Common Data Environments (CDEs);
2. **For communication and dissemination**, as visual representations, such as photographs, drawings and diagrams in books, magazines, and websites.

These forms create a tension between structured but often inaccessible project data and accessible but weakly machine-readable public representations.

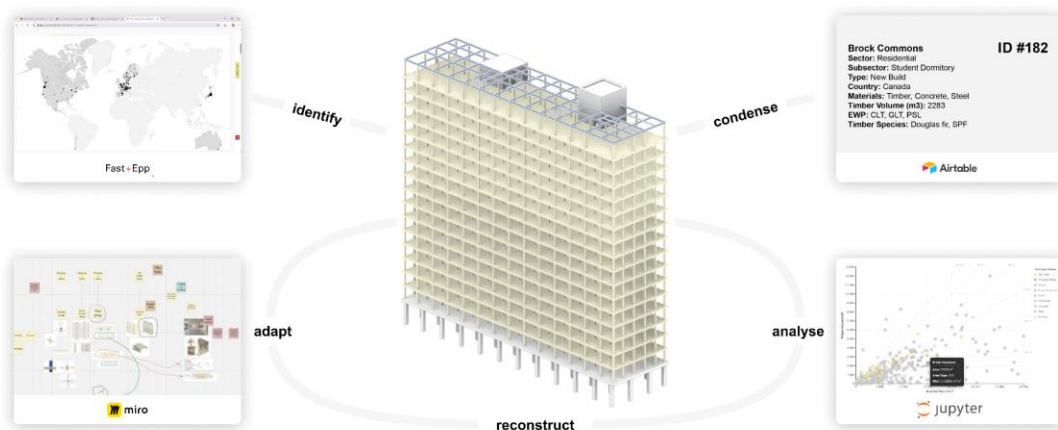


Fig. 1 Exemplary workflow of architectural research data collection.

As a result, architectural research often reconstructs data manually from public visual sources (Winkler, 2026). Such reconstructions are rather approximations and abstractions reduced to information relevant for a specific research context than holistic digital twins. The generated 3D artefacts are not merely a central output of the research process, but a result of analytical engagement with the building and thus part of the methodological process (Fig. 1).

Against this background, this paper asks what requirements derived metadata and data must fulfil to support the FAIR publication of research-specific building data. It distinguishes between dataset-level FAIRness, addressed by repositories through persistent identifiers, metadata, licences, and machine-readable records, and data-level FAIRness, concerning the internal structure and documentation of the data itself.

The Architectural Research Data Management-Pipeline (ARDMP) addresses both levels: It connects tabular data (e.g. Airtable) and transforms it into RDF graphs with stable identifiers for entities, sources, files, drawings, models, and relationships. Shapes Constraint Language (SHACL) based Application Profiles provide descriptions and constraints making the schema explicit and understandable. Where possible, ontologies, controlled vocabularies and authority data enrich entities such as locations, organisations, building types and materials. RO-Crate is used as an export container to organise files, metadata, and provenance information as a FAIR Digital Object prepared for the publication in NFDI4ING Data Ingest Service (Fig. 2). A central concern is the documentation of abstraction levels, uncertainty, and the reasoning behind classifications and modelling decisions. Since such metadata emerges iteratively, pragmatically, and sometimes incompletely, it can be understood as *metadata-as-process* (Edwards, 2011). FAIR publication therefore requires not only formal structure, but transparent documentation practices of what the data represents, omits, and under which assumptions it was produced.

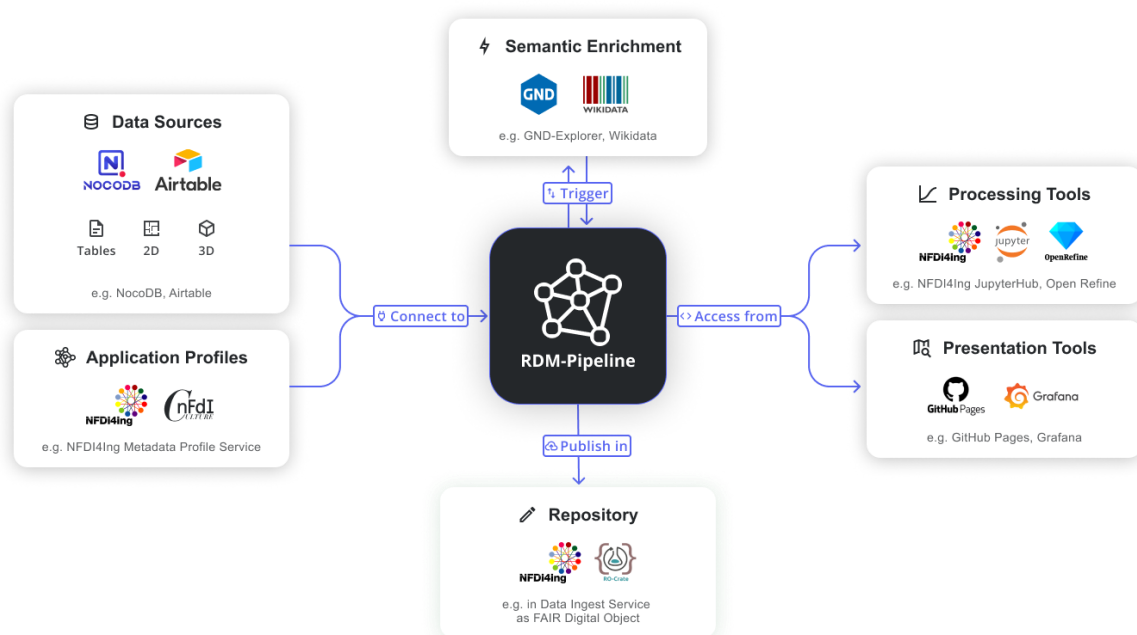


Fig. 2 Conceptual diagram of the approach with the architectural research data management pipeline

Based on a collection of multi-storey timber buildings (Svatoš-Ražnjević, 2022), the paper discusses how this approach mediates between human- and machine-readable representations. It further reflects on the proposed workflow in relation to historical research in digital heritage, while also considering how established documentation practices from 3D reconstruction, especially concerning sources, assumptions, uncertainties, and decision processes (Grellert, 2023), can inform the creation of datasets on contemporary architectural works. In this way, shared workflows are explored as a means to enhance transparency, comparability, and reuse, highlighting parallels and synergies across the domains.

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